

Problem Statement ID- IOT-03

Problem Statement Title- Real-Time Cold Chain Integrity Monitoring for Vaccine Logistics

Title - AI Enhanced Vaccination Kiosk

Team Name - MethXAI

Team Leader Name - ANIKET KUMAR

Members Name - RAVI PRAKASH RAI
VANSI
NIKHIL KUMAR



“Vaccines can lose potency silently in transit—we catch the excursion while the shipment can still be acted on.”

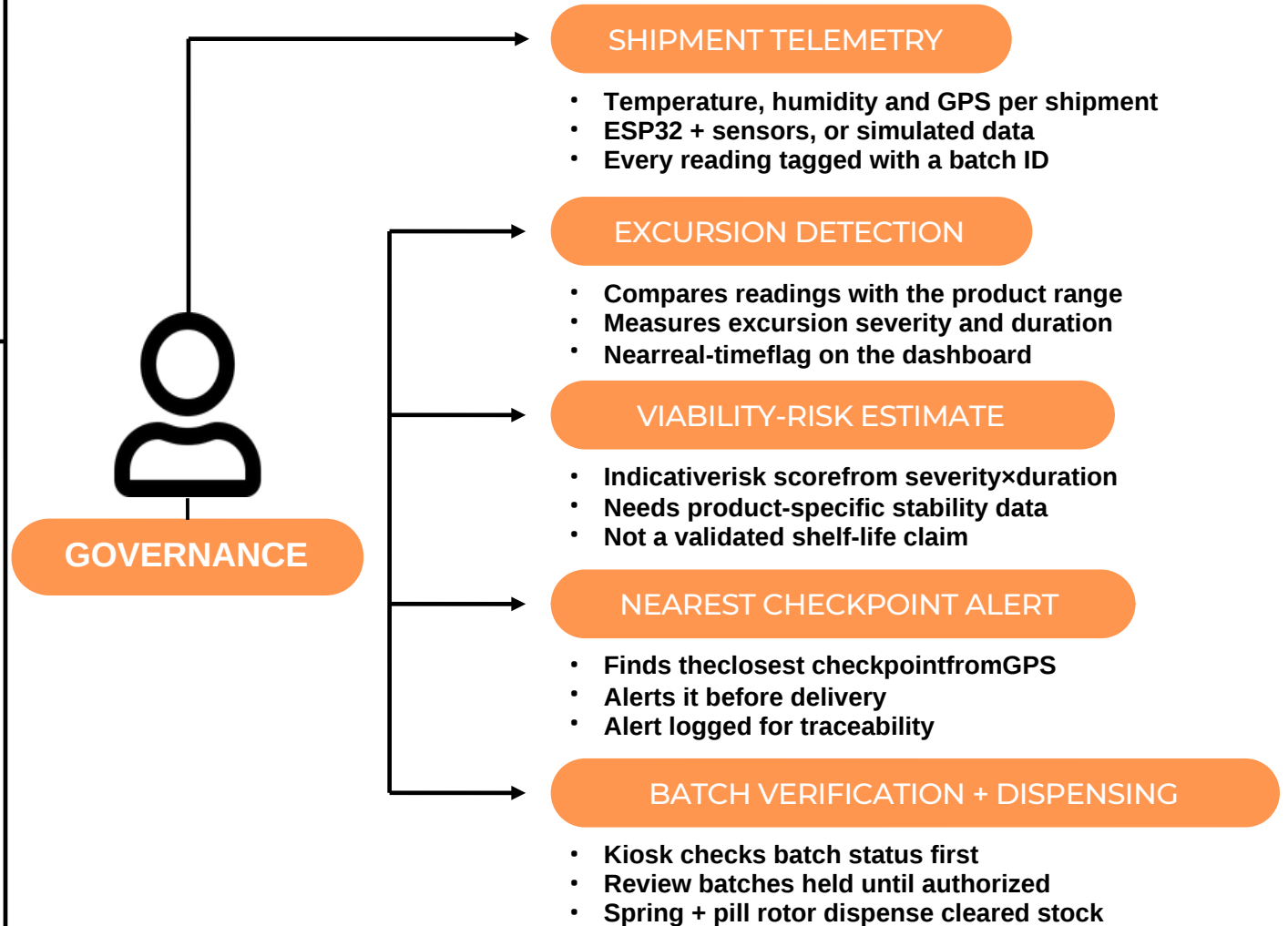
► PROPOSED SOLUTION

MethXAI's an IoT- enabled cold-chain monitoring and intelligent medicine distribution system. It tracks temperature, humidity and GPS of vaccine shipments, flags excursions in near real time and alerts the nearest checkpoint.

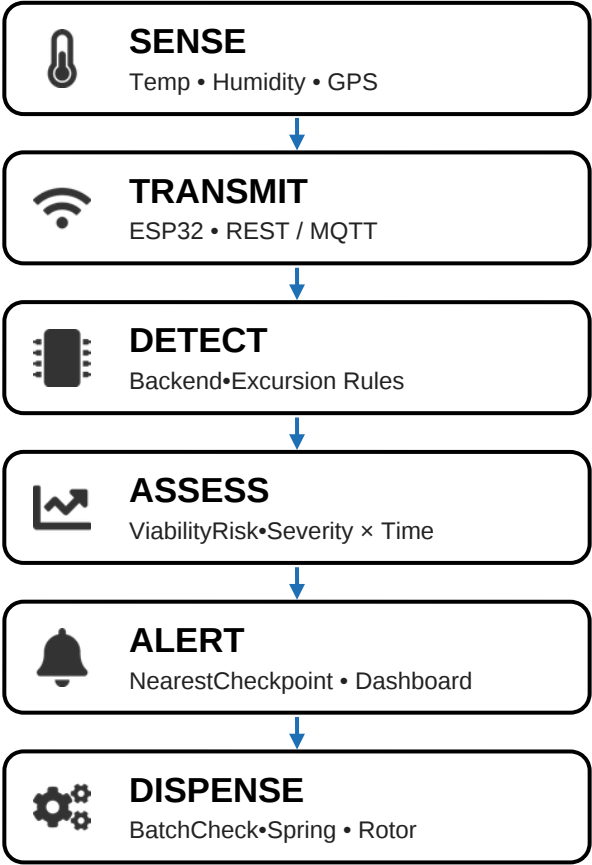
► INNOVATION & UNIQUENESS

- Automatic nearest-checkpoint alerting
- Predictive excursion detection and dynamic vaccine viability estimation
- Risk aware intervention
- Multi - sensor risk correlation
- Offline and connectivity-gap detection

► SYSTEM WORKFLOW



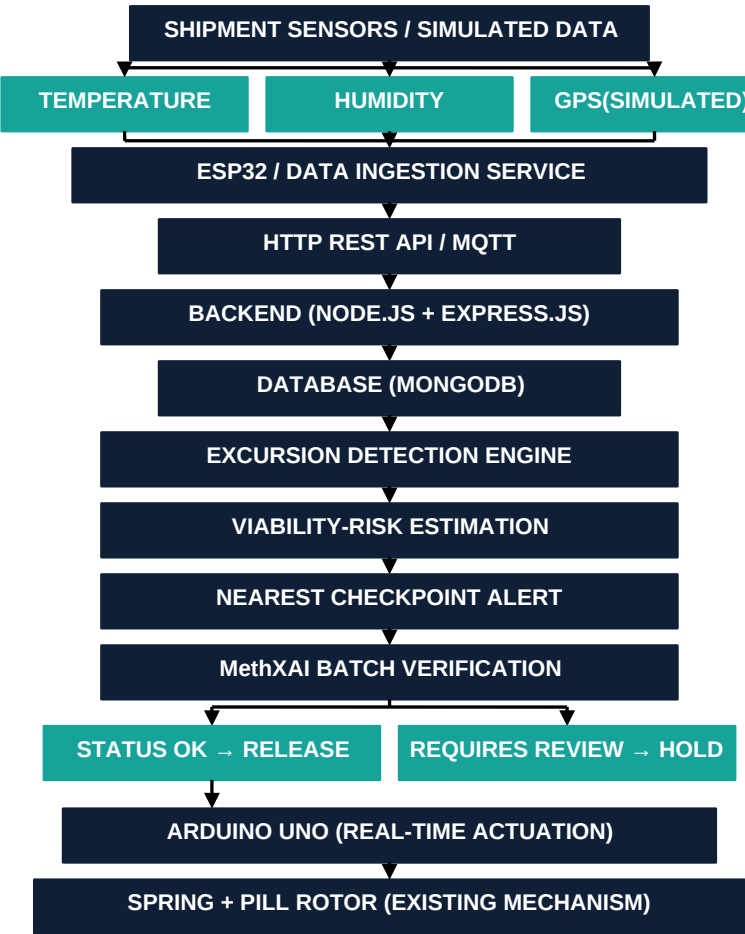
METHODOLOGY



CORE WORKFLOW

SENSE → DETECT → ASSESS → ALERT → VERIFY → DISPENSE

FLOW CHART



- SHIPMENT SENSING

 - DS18B20 — Temperature (prototype)
 - SHT31 — Temperature & Humidity
 - Simulated GPS (module optional)
- CONNECTIVITY & BACKEND

 - ESP32 — Data Acquisition, Wi-Fi
 - HTTP REST API or MQTT telemetry
 - Node.js + Express.js + MongoDB
- EXCURSION & RISK ENGINE

 - Rule-based excursion detection
 - Severity + duration → viability risk
 - Alert to nearest checkpoint
- AUTOMATED DISPENSING

 - Batch status check at the kiosk
 - Arduino UNO — Motor Controller
 - Spring mechanism + Pill Rotor
- DIGITAL RECORD

 - Shipment & batch traceability
 - Excursion / alert log
 - Dispense and hold events

TECHNOLOGY STACK

Embedded:

ESP32 • ArduinoUNO • DS18B20 • SHT31 • Peltier • Fan • Simulated GPS • UART • I²C • PWM

Software:

React.js • Vite • Node.js • Express.js • Socket.IO • MongoDB • REST API • MQTT

Data:

Kaggle cold-chain datasets (simulated replay) • Rule-based risk engine

1. FEASIBILITY :

Existing Components

- ESP32 + Arduino UNO controllers
- Spring + pill rotor dispenser (unchanged)
- React + Node.js + MongoDB stack

Prototype Approach

- DS18B20 / SHT31 on ESP32 for readings
 - Simulated GPS + dataset replay
 - Rule-based excursion and risk logic
 - Only new dispenser logic: batch check
- **Technically achievable with currently available components. No refrigeration hardware added.**

2. RISKS → CONTROLS :

- Sensor failure** → Flag UNKNOWN, hold batch
- Network loss** → ESP32 buffers, then backfills
- Excursion** → Batch set to REQUIRES REVIEW
- Unvalidated estimate** → Labelled indicative only
- Temperature-only data** → Never declared "safe"
- FAIL-SAFE PRINCIPLE**
- Uncertainty** → **NO DISPENSE**

3. USE CASES :

- Vaccine transport** → In-transit excursion alerts
- Checkpoint / cold store** → Pre-delivery warning
- District store** → Batch check before issue
- Rural centre kiosk** → Verified last-mile dispensing

4. VIABILITY :

OPERATIONAL

- Live shipment monitoring
- Checkpoint alerts before delivery
- Digital audit trail per batch

COST

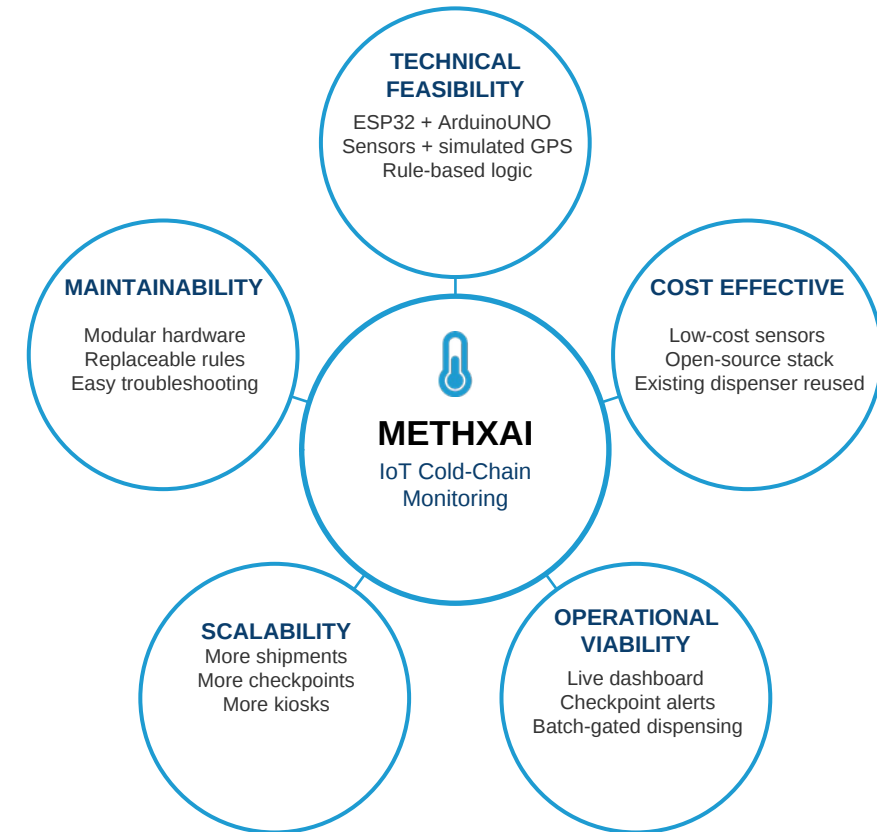
- Low-cost sensors, open-source software
- Existing dispenser reused as is

SCALABILITY

- 1 Shipment → Pilot → Checkpoint Network
- Cold stores | Hospitals | Health Centres | Kiosks

5. SCALABILITY PATHWAY :

- 1. Simulated Demo** — ESP32 sensors + dataset replay
- 2. Pilot Shipment** — real sensors on one route
- 3. Checkpoint Network** — multiple checkpoints, alert routing
- 4. Validated Rules** — product stability data, expert review
- 5. Connected Kiosks** — batch-gated dispensing across centres



ALTERNATIVES & OUR EDGE

ALTERNATIVES

Ice-lined refrigerators

- High initial cost
- Heavy & bulky
- Risk of vaccine freezing (needs +2°C to 8°C)

VVM (Vaccine Vial Models)

- Do not provide live GPS tracking, humidity data or alerts during transit.
- They do not detect freezing damage.

OUR APPROACH

PRECISION

Temp • Humidity • GPS



DECISION MAKING

Excursion rules • Viability risk



CORRECTIVE ACTIONS

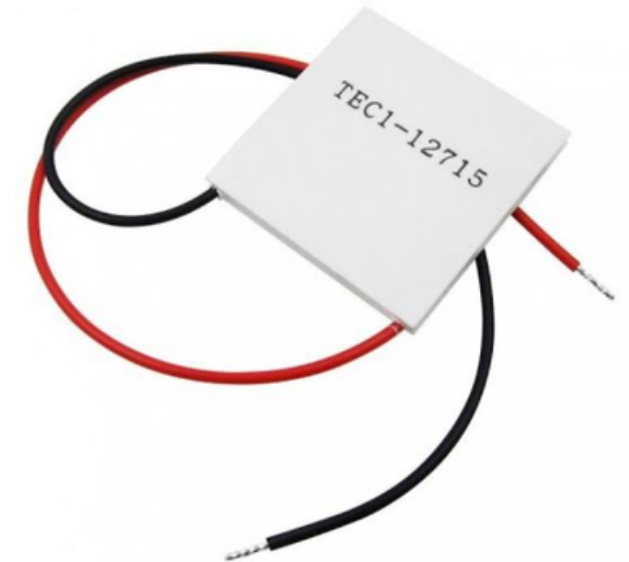
Use of Peltier Module & AHU



ALL IN ONE SYSTEM

PLUS POINTS

- ➔ EARLIER INTERVENTIONS
- ➔ RELEVANT WARNINGS
- ➔ FEWER MISLEADING ASSESSMENTS



KEY TAKEAWAY

Existing options only store or alert. MethXAI adds precision, decision making and corrective action in one system.

IMPACTS

MethXAI turns a conventional medicine dispenser into a **cold-chain-aware distribution endpoint** — combining **real-time shipment monitoring**, **excursion alerts**, **viability-risk estimates** and **batch-checked dispensing**

BENEFITS

Social



Verified stock for hospitals, health centres and underserved/rural locations

Digital



Live dashboard, alert log and batch records in one place

Traceability



Shipment-to-batch records support audit and investigation

Resource Efficiency



Earlier excursion alerts can reduce avoidable vaccine wastage

Healthcare Safety



Batch status check + hold on review batches before dispensing

Environmental



- Less wasted vaccine → fewer repeat shipments
- Less paper → digital logs
- Reuse of existing dispenser hardware

OUR PROTOTYPE

DEMO FLOW (SIMULATED SHIPMENT)

1. Simulate a vaccine shipment moving toward a checkpoint
2. Display temperature, humidity and GPS data
3. Trigger a temperature excursion
4. Apply Temperature handling : Peltier Cooling Module
5. Monitor Humidity : Fan/air Circulation + Air Handling unit
6. Calculate a labelled viability-risk estimate
7. Alert the nearest relevant checkpoint
8. Mark the batch as requiring review
9. Dispensing held until authorization

ESP32 → ARDUINO UNO → SPRING + PILL ROTOR

Existing hardware: ESP32, Arduino UNO, spring dispenser, pill rotor

To build: DS18B20 / SHT31 readings, backend, dashboard, batch check

Simulated: shipment route, GPS, dataset replay, risk rules

Cold-Chain Monitoring + Batch-Checked Dispensing

DATASETS AND REFERENCES

•Dataset 1 — Cold ChainShipmentSilent Failure(Kaggle):

Planned use: replay as simulated shipment telemetry

Link- <https://www.kaggle.com/skarin/cold-chain-shipment-silent-failure-dataset>

•Dataset 2 — Vaccine Distribution with Temperature Logging (Kaggle):

Planned use: temperature-log replay and excursion-rule testing

Link- <https://www.kaggle.com/manankhanna0/vaccine-distribution-with-temperature-logging>

•Viability estimation basis:

Needs product-specific stability data and validated rules or models. The demo score is an indicative risk label, not a shelf-life claim.

•Existing hardware base:

MethXAI V1 dispensing prototype: ESP32 + Arduino UNO with spring mechanism and pill rotor.

MethXAI: from the first sensor reading to the last dispensed dose, the cold-chain record decides what leaves the kiosk.